

Tejal Anavekar

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

Aug 2024 – May 2026

- Relevant skills: Analysis of Algorithm, Database management, Applied Natural Language Processing, Machine Learning for Data Science, Information Retrieval

Fr.C.Rodrigues Institute of Technology

Mumbai, India

Bachelor of Engineering in Information Technology

July 2020 – June 2024

- Relevant skills: Operating Systems, Computer Networks, Machine Learning, Artificial Intelligence, Data Science, Data Mining, Big Data

TECHNICAL SKILLS

Programming: Python, C, C#, HTML, CSS, Javascript

Systems & OS: Linux/Unix, Shell Scripting, TCP/IP, HTTP/HTTPS

Framework: ASP.NET Core, .NET, ReactJs, NodeJs, ExpressJS, APIs

Libraries: PyTorch, Scikit-learn, Keras, OpenCV, TensorFlow, Bootstrap, Pandas, NumPy, Seaborn, Matplotlib

Databases & Cloud : SQL, Mongo DB, PostgreSQL, Firebase, AWS (EC2, S3), GCP

EXPERIENCE

Software Developer

Sept 2024 – Present

USC Institute for Creative Technologies

Los Angeles, CA

- Designed and implemented backend services in **C#** to process and transform large-scale production data, optimizing SQL queries and improving system throughput by **40%**
- Debugged and resolved cross-service frontend-backend failures using JavaScript, improving production system reliability.
- Built and supported production monitoring dashboards and backend data services, integrating **real-time data streams** to track critical financial KPIs

Data Engineer Intern

June 2025 – Aug 2025

Meta Platforms, Inc

Burlingame, CA

- Designed and implemented **4** scalable pipelines for distributed data processing and handling **billions of records** to power real-time analytics and AI readiness dashboards in production environments
- Built automated validation and processing frameworks in Python to ensure scalability, fault tolerance, and reliable production data workflows
- Refactored and optimized computation logic to reduce processing overhead and improve performance, leading to adoption across **3+** cross-functional engineering teams

AI Engineer

June 2022 – Apr 2023

Tata Institute of Fundamental Research

Mumbai, India

- Applied advanced image segmentation and **SIFT-based feature extraction**, boosting classification precision by **25%** and enabling more precise data interpretation
- Trained and deployed **Convolutional Neural Networks** (ResNet, VGG variants) to automate image classification tasks, improving accuracy and cutting processing time by 30%
- Designed modular **2** preprocessing pipelines to standardize image resolution and brightness for robust ML training across diverse datasets and imaging conditions

PROJECTS | PUBLICATIONS

Budget Buddy | *MERN, LangChain*

Nov 2025 – Present

- Built a full-stack MERN expense management system with RESTful APIs, MongoDB schema, and AI-powered financial insights, developed an interactive dashboard with real-time monthly spending charts and category-based segmentation, improving users' budget tracking efficiency by 35%

Query Bot | *NLP, BERT, LSTM, Pytorch*

Jan 2025 – Apr 2025

- Led the development of a custom seq2seq model combining **BERT encoders and LSTM decoders** with schema-aware attention for translating natural language questions to SQL queries. Achieved 91.8% execution accuracy and 82.2% exact match on Spider dataset

GeoForecast - Weather App | *MERN Stack, Flask, Swift, GCP*

Aug 2024 – Dec 2024

- Designed and deployed a **modular microservices-based backend architecture** using Node.js and MongoDB, hosted on GCP with scalable API integrations. Integrated Google and Tomorrow.io APIs for geolocation, weather prediction and daily forecasts, achieving 93% location accuracy and increasing daily retention by 35%

Smart Surveillance System for Intrusion Detection | *SSD, YOLO, Open CV, Tensorflow*

July 2023 – Apr 2024

- Devised and Implemented AI-powered smart surveillance system utilizing **SSD and YOLO** algorithms for real-time leopard detection across 500 meters, achieving 92% detection accuracy with false positive reduction of 30%
<https://rjpn.org/jetnr/papers/JETNR2404011.pdf>